

Introduction to Fractional Calculus

Example. Compute $4!$

Solution. $4! = 1 \cdot 2 \cdot 3 \cdot 4 = 24$

Example. Using the fact that

$$\Gamma(2) = 1! = 1, \quad \Gamma(3) = 2! = 2, \quad \Gamma(5/2) = \frac{3}{4}\sqrt{\pi}$$

evaluate

(a) ${}_0I_x^{1/2}[x]$

(b) ${}_0I_x^{1/2}[x^{3/2}]$

Solution. First,

$$\Gamma(5/2) = \Gamma(3/2 + 1) = \frac{3}{2} \Gamma(3/2) = \frac{3}{2} \cdot \frac{1}{2} \Gamma(1/2) = \frac{3}{4}\sqrt{\pi},$$

(a) Using the power rule for the Riemann–Liouville fractional derivative we get

$${}_0I_x^{1/2}[x] = \frac{\Gamma(2)}{\Gamma(2 + 1/2)} x^{3/2} = \frac{\Gamma(2)}{\Gamma(5/2)} x^{3/2} = \frac{1}{\frac{3}{4}\sqrt{\pi}} x^{3/2} = \frac{4}{3\sqrt{\pi}} x^{3/2}.$$

(b) Next,

$${}_0I_x^{1/2}[x^{3/2}] = \frac{\Gamma(5/2)}{\Gamma(3)} x^2 = \frac{\frac{3}{4}\sqrt{\pi}}{2} x^2.$$

Thus,

$$\begin{aligned} {}_0I_x^{1/2} {}_0I_x^{1/2}[x] &= {}_0I_x^{1/2} \left[\frac{4}{3\sqrt{\pi}} x^{3/2} \right] = \frac{4}{3\sqrt{\pi}} {}_0I_x^{1/2}[x^{3/2}] \\ &= \frac{4}{3\sqrt{\pi}} \cdot \frac{\frac{3}{4}\sqrt{\pi}}{2} x^2 \\ &= \frac{x^2}{2} = {}_0I_x^1[x] = \int_0^x t \, dt. \end{aligned}$$

Example. Using the fact that

$$\Gamma\left(\frac{7}{2}\right) = \frac{15\sqrt{\pi}}{8},$$

compute the fractional derivative of x^2 .

Solution. Computing the half derivative of x^2 we obtain

$${}_0D_x^{1/2}[x^2] = \frac{d}{dx}({}_0I_x^{1/2}[x^2]) = \frac{d}{dx}\left(\frac{\Gamma(3)}{\Gamma(7/2)}x^{5/2}\right) = \frac{5\Gamma(3)}{2\Gamma(7/2)}x^{3/2} = \frac{5}{\frac{15\sqrt{\pi}}{8}}x^{3/2} = \frac{8x^{3/2}}{3\sqrt{\pi}}.$$

Differentiating the result obtained

$${}_0D_x^{1/2}\left[\frac{8x^{3/2}}{3\sqrt{\pi}}\right] = \frac{8}{3\sqrt{\pi}}\frac{d}{dx}({}_0I_x^{1/2}[x^{3/2}]) = \frac{8}{3\sqrt{\pi}} \cdot \frac{\frac{3}{4}\sqrt{\pi}}{2}\frac{d}{dx}(x^2) = 2x.$$

As expected,

$${}_0D_x^{1/2} \circ {}_0D_x^{1/2}[x^2] = {}_0D_x^1[x^2] = \frac{d(x^2)}{dx}.$$